EPSR-test-ex27 changeato: keyword navigation and molecule removal

Program(s): changeato | chained passes: 1 | verification: ALL IDENTICAL (1 pass)

What this case tests

Exercises the remaining interactive paths of `changeato` [1]: removing a number of molecules of chosen types from a configuration, moving through the keyword list (`d`/`u`, bare values, `atom n`), `help`, `default`, unknown commands, relabelling all atoms of a type, and the different exits (`q`, `bye`, save at the final prompt).

Code exercised

Fortran (original, fortran_src/):

- `changeato.f`: `changeato`; `process_symbols.f`: `process_symbols`, `process_commands` (`d`, `u`, `help`, `default`, `save`, `q`, `bye`), `check_symbol`, `change_symbol_changeato`, `write_symbols_changeato`
- `process_symbols.f`: `remove_molecule_types`, `remove_molecule`, `cleanup_types`, `change_label`, `save_files_changeato`
- `read_and_save_files.f`: `save_files`; `ato_read_and_save.f`: `minitc`, `moloutc`

Rust (port, rust/epsr-core/src/):

- `tools/changeato.rs`: `run`, `process_symbols`, `process_commands`, `check_symbol`, `change_symbol_changeato`, `write_symbols_changeato`
- `tools/changeato.rs`: `remove_molecule_types`, `remove_molecule`, `cleanup_types`, `change_label`, `save_files`; `ato.rs`: `minitc`, `moloutc`

Physical system

`MixD.ato` is the EPSR-test mixture of 200 DMSO and 400 water molecules (3200 atoms, 300 K, cubic box 33.1213 Å); 3 DMSO molecules are removed from the start of the file and 395 water molecules, leaving 202 molecules (197 DMSO + 5 water) in the same box, and several parameters are changed through keyword navigation (e.g. `sigma 3.4` for atom type 1). `WaterE.ato` is the single EPSR-test water molecule (box 14.422 Å) with SPC/E-type parameters [2]; the H type is relabelled `HX`, the epsilon of O set to 0.7 kJ/mol, all H atoms (fraction 1.0) relabelled `HY` and the temperature set to 250 K.

Inputs

`MixD.ato` = `EPSR-test/200DMSO400Water.ato`, `WaterE.ato` = `EPSR-test/WaterMolecule.ato`; commands in `caseD.stdin` and `caseE.stdin`, including a blank answer, a non-numeric value, `atom 99` (clamped to the last type) and the unknown command `unknowncmd`.

How the accuracy was verified

`commands.txt` (2 program calls, run once) was run with the Fortran reference build (`rust/tools/build_fortran_reference.sh`: the original sources, gfortran -O2 -fno-tree-vectorize, single thread) and with the Rust `epsr-rs` on identical copies of the inputs, and every file written was compared byte for byte (`rust/tools/run_ex_test.sh`, `compare_outputs.py`; no masking except where noted below). The Fortran output is stored in `fortran_reference.tar.gz`. Result of the final regression run of the complete port: every file identical in 1 pass.
One pass of two `changeato` calls; compared files: `MixD.ato` (header: 202 molecules) and `WaterE.ato` (header: 250 K).

Notes

When molecules are removed from the start of the file, the Fortran does not renumber the bond partners and rotation-group atoms of the molecules that move down; this is reproduced exactly (README).

References

[1] A. K. Soper, "Empirical Potential Structure Refinement - EPSRshell: a user's guide", Technical Report RAL-TR-2011-012, STFC Rutherford Appleton Laboratory (2011).
[2] H. J. C. Berendsen, J. R. Grigera and T. P. Straatsma, "The missing term in effective pair potentials", J. Phys. Chem. 91, 6269-6271 (1987).
[3] GNU Fortran (gfortran) 15.2, GCC, https://gcc.gnu.org/fortran/ - reference build with -O2 -fno-tree-vectorize.